

Prasanna J

Data Engineer

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Summary

Data Engineer with 2 years of experience building large-scale AWS-based data pipelines processing 350M+ records. Specialized in Spark (EMR), Airflow orchestration, distributed transformations, and data quality frameworks. Strong focus on scalability, cost optimization, incremental processing, and production reliability in cloud-native environments.

Skills

Cloud & Infrastructure: S3, EMR, Glue, EC2, Lambda, IAM, CloudWatch, Secrets Manager

Infrastructure as Code & Automation: Terraform

CI/CD & Automation: GitLab CI/CD, Git, Pipeline Automation

Data & Engineering: ETL/ELT Pipelines, Batch Processing, Distributed Data Systems, Data Modeling (Star Schema, Dimensional Modeling), Apache Spark, PySpark, AWS EMR, Airflow, Glue, Snowflake, Incremental Processing, Schema Evolution, Partitioning, Performance Tuning, Data Validation, Data Warehousing

Programming: Python, SQL (Advanced CTEs, Window Functions)

Work Experience

Data Engineer, Lightcast

Nov 2025 – Present

- Designed and deployed distributed ETL pipelines using Apache Spark (PySpark) on AWS EMR processing datasets from 5+ external data providers and handling hundreds of millions of records.
- Reduced Spark job runtime by 30–40% by optimizing partitioning strategy, caching frequently accessed datasets, and tuning executor memory and shuffle configurations.
- Built modular data pipeline stages integrated with GitLab CI/CD enabling automated deployment, validation checks, and reliable production releases.
- Architected scalable data lake storage using AWS S3 with optimized Parquet formats and partitioning strategies improving query performance in AWS Athena.
- Implemented automated data validation and lineage tracking improving pipeline reliability and reducing production data issues.
- Processed 350M+ records across distributed Spark pipelines ensuring scalable and reliable batch processing.

AWS Data Engineer, Gana Outsourcing

July 2024 – Oct 2025

- Built an end-to-end AWS data pipeline using Medallion Architecture (Bronze, Silver, Gold) to transform and curate large-scale operational datasets.
- Ingested data from Oracle databases via AWS Glue JDBC connectors and stored raw data in S3 as partitioned Parquet datasets enabling scalable data lake storage.
- Reduced data pipeline processing time from 3 hours to 30 minutes by implementing incremental processing and partition-based optimizations.
- Developed transformation logic in PySpark to standardize and cleanse datasets including deduplication, normalization, and schema validation.
- Implemented incremental data loading with watermarking and MERGE operations to load curated datasets into Snowflake data warehouse.
- Automated workflow orchestration using AWS Managed Airflow ensuring reliable scheduling, monitoring, and failure recovery.

Education

Anna University Regional Campus, Trichy

Aug 2020 – Jun 2024

Bachelor of Engineering (B.E.) in Civil Engineering

GPA: 7.6 / 10